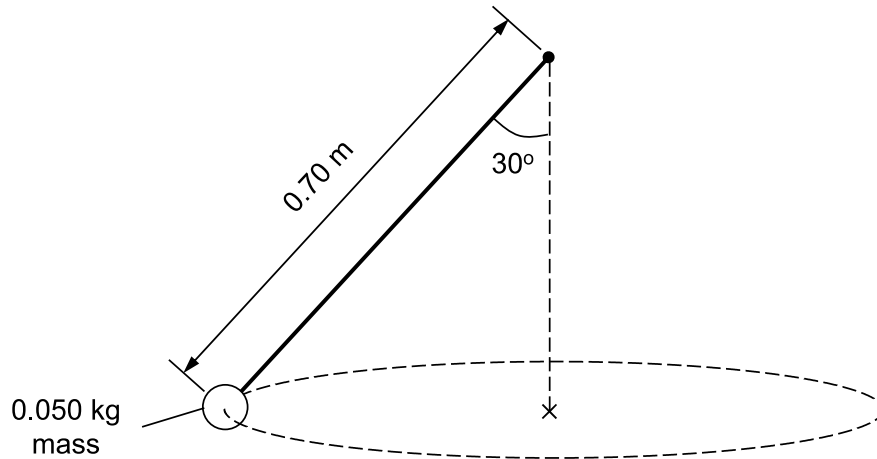


10. A mass of 0.050 kg is attached to one end of a piece of elastic cord of unstretched length 0.50 m. The elastic cord obeys Hooke's Law and its force constant is 40 N m⁻¹.

The mass is rotated in a horizontal circular path as shown in the diagram.



The length of the cord is 0.70 m.

What is the speed of the mass?

- A** 5.3 m s⁻¹ **B** 7.5 m s⁻¹ **C** 9.2 m s⁻¹ **D** 28 m s⁻¹

Answer: A

$$\text{force on mass by cord} = 40 \square (0.70 - 0.50) = 8 \text{ N}$$

$$\text{radius of circular path} = 0.70 \sin 30^\circ$$

since horizontal component of force by cord provides centripetal force,

$$8 \sin 30^\circ = \frac{0.050 v^2}{0.7 \sin 30^\circ}$$

$$\square v = \sqrt{\frac{8 \sin 30^\circ \square 0.7 \sin 30^\circ}{0.050}} = 5.3 \text{ m s}^{-1}$$

